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 SURVEY

A Review on Security Challenges in V2X Communications Technology for VANETs

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ABSTRACT The Vehicular Ad-hoc Network (VANET) is emerging as a new networking/communication model that offers remarkable approaches for controlling and managing vehicular traffic. In this scenario, Virtual traffic light (VTL) techniques, advanced navigation support, and fleet management facilities attempt to handle traffic management and control concerns by leveraging vehicular network communication paradigms. The most important communication models in vehicular networks are characterized as vehicle-to-everything (V2X) communications technology. In V2X communications, security issues, and related countermeasures are crucial, as they are in other wireless technologies. Our survey provides a systematic overview of various research directions by highlighting and investigating the most critical V2X security issues and related countermeasures. First, we discuss the points mentioned in the previous papers, including the architecture, main characteristics, and communication models of the V2X networks. Furthermore, some of the most significant applications of these mission-critical networks reported in the published papers and the main security challenges and requirements will be highlighted. We also arrange corresponding (existing) resources, to investigate different solutions for securing V2X communications. Finally, a few suggestions for future research related to V2X security challenges are presented. In general, this article reviews and summarizes the studies of the last ten years. This survey will help make it easier for scholars studying V2X communication security and related fields to keep track of the academic frontier, by offering a sufficiently clear landscape and a roadmap.

INDEX TERMS VANET, V2X communications, security, privacy, cyberattack, security solutions, cryptography, machine learning.

I. INTRODUCTION

A. OVERVIEW

VANETs are gaining extreme interest in the research world because of their focus on safety and convenience for both passengers and drivers [1], [2]. To enhance the safety of roads and provide passengers with enhanced driving-related and entertainment services, VANETs connect vehicles and road infrastructures. Vehicles are equipped with wireless sensors, location equipment, cruise control, navigation facilities, and related maps. On-Board Units (OBUs) connect vehicles to Road-Side Units (RSUs) to exchange information between

vehicles and between vehicles and RSUs [3], [4]. Together with OBUs and RSUs, a trusted authority component is also needed in the VANET architectures, providing centralized network management and supervision [5].

A large variety of algorithms and facilities have been proposed for VANETs. VTL is one of the most well-known traffic optimization techniques for enhancing vehicle movement on roads and intersections. Starting from proper vehicular network communication models, VTL approaches for VANETs attempt to handle traffic control challenges. These communication models can be categorized into: Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Network (V2N), and Vehicle-to-Pedestrian (V2P) communications. These four possibilities together are often

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referred to as vehicle-to-everything (V2X) communications technology [6], which are depicted in Figure 1. V2X communications are able to improve safety, passenger entertainment, and vehicular traffic flow control with V2X communications facilities leveraging existing vehicle sensing capabilities, and also handle a wide range of use cases, like forward collision warning, do-not-pass warning, queue warning, curve speed warning, optimal speed advice, and parking discovery. Several wireless communication technologies, such as Dedicated Short-Range Communications (DSRC) and cellular network technologies, are utilized in V2X communications to exchange data packets between infrastructure, vehicles, and pedestrians to enable various new use cases (e.g., protecting vulnerable road users, providing real-time situational awareness in lane change/merging, supporting speed harmonization, autonomous driving, and platooning).

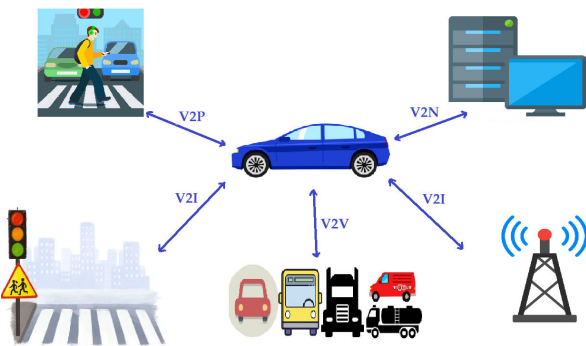


FIGURE 1. V2X communications scenarios.

Future Intelligent Transport Systems (ITS) depend on the V2X communication technology that forms an essential part of their structure. Motor vehicles have dramatically shifted over the past decade from simple mechanical devices to extremely complex systems incorporating sophisticated control, management, and decision technologies. In today's cars, sensors assess various factors, enhancing the driver and passenger experience while also improving safety. However, there are still numerous other important issues with the transportation system. In recent years, there has been a significant expansion of urban regions and a surge in motor vehicle traffic in big cities. As a result, traffic problems and vehicle accidents are increasing on city streets and highways, causing severe socio-economic problems. New data from the World Health Organization reveals that 1.19 million people die yearly in vehicle crashes [7] which result in the leading cause of death for children and young adults. More than 50% fatalities involve vulnerable road users, including pedestrians, cyclists, and motorcyclists. Of course, injuries and fatalities may further increase with the number of travelers, without a corresponding improvement in safety. The United Nations Decade of Action for Road Safety 2021–2030 indicates among its main objectives the reduction of road traffic deaths and injuries by at least 50% by 2030 [7].

Therefore, government organizations, research institutions, and vehicle manufacturers made significant attempts to address traffic safety issues by equipping transportation infrastructure and vehicles with communication abilities. In particular, V2X communications are responsible for boosting the cooperation between transportation infrastructure, vehicles, and pedestrians, which is expected to significantly decrease roadway crashes, leading to secure and safer ground transport systems. Therefore, the goal of V2X is to allow applications that could significantly enhance the quality of transportation systems and the average performance of vehicular networks [8], [9]. Vehicles are getting cluttered with a variety of data and communication technologies as a way of improving interconnection among different V2X components. However, using such units and technologies for massive data collection and distribution presents several concerns, most notably in terms of privacy and security. An attacker may use the system's vulnerability to obtain access to and subsequently control vehicles, potentially resulting in hazardous driving circumstances and life-threatening accidents. From a categorization standpoint, vulnerabilities in V2X and vehicular communications result in four distinct vehicular security concerns, as follows:

1) UNPREDICTABLE THREATS AND ATTACK SCENARIOS

Remote communication technology, vehicle database, roadside infrastructure, and vehicle parts can all be used to infiltrate a V2X architecture. Automobile manufacturers will have great difficulties in predicting where hackers will hit next because new techniques are constantly created. An original product that has been left unprotected and unsecured can give hackers new points of entry into a V2X system.

2) LIMITED CONNECTIVITY

Although vehicles' external connectivity is expanding, most vehicles cannot timely receive over-the-air upgrades for their onboard systems, allowing them to be constantly protected against the newest security assaults and menaces. Over-the-air updates are increasingly standardized, but disconnected cars are still vulnerable due to incomplete or missing updates.

3) LIMITED COMPUTATIONAL PERFORMANCE

As one of the crucial concerns in V2X communications, a vehicle's computing power is typically limited and significantly lower when compared to a modern computer, tablet, or smartphone. Moreover, in consideration that automobiles have a prolonged lifespan and their obsolescence is significantly longer than the one characterizing common computing equipment, sometimes their computing equipment cannot be updated. Therefore, vehicles are more vulnerable to hacking than computers due to their processing and computational deficit [10]. Some vehicular security solutions cannot be executed due to the high overhead associated with their limited computational capacity and performance. Furthermore, they are usually designed with or without limited security concerns in mind.

4) SIGNIFICANT RISKS TO THE SAFETY OF DRIVERS AND PASSENGERS

Vehicle faults that can put drivers, passengers, and pedestrians' lives in danger can occur even when only a tiny number of sensors have been compromised, or a small number of illegitimate messages have been delivered. Motivated by the importance of the above-mentioned issues, in this survey study, we focus on cybersecurity issues, security challenges, and their solutions in V2X communications, whose awareness can be extremely useful in facing many security problems in vehicular networks. Particularly, we provide a comprehensive and systematic overview of various research directions by highlighting and investigating the most critical V2X security issues and related countermeasures.

B. EXISTING SURVEYS AND LIMITATIONS

Various surveys have been produced on V2X systems in the past years. Only a tiny fraction of them have concentrated on security challenges. The authors in [11] investigate V2X security issues in cellular networks and the V2X authentication methods that have been proposed thus far. However, recent developments based on machine learning are not considered. The authors of [12] and [13] highlight security vulnerabilities, risks, and issues that may be utilized to motivate the development of a plan to tackle cybersecurity-related problems in V2X. However, the problems and mitigation strategies they discussed have become outdated as a result of the rise of modern advancements (for example, machine learning-based strategies, fifth and sixth-generation wireless technologies (5G and 6G), and blockchain) that may accelerate the development of more secure solutions. The authors of [14] and [15] focus on identifying intrusions and misbehavior in vehicular networks.

The work in [16] presents an examination of the security characteristics of 5G, including standards, challenges, and solutions, as well as their applicability to V2X communication and vehicular networks. However, recent developments based on machine learning are not considered in this research. In [17], the authors present a security analysis related to vehicular networks, paying particular attention on anonymous authentication techniques that safeguard vehicle users' privacy, as well as trust management methods. In [18], the authors analyze the vulnerabilities and threats associated with V2X and various traditional security mechanisms. The authors of [19] investigate the state-of-the-art security mechanisms for V2X communications. However, this study did not address new mechanisms that have emerged recently (like 5G-V2X security technologies and defense mechanisms based on machine learning). Moreover, it lacks solution methods for attacks against safety-critical applications. The research in [20] investigates the main characteristics of V2X and concentrates on the standardization methods used for communication technologies. Moreover, the authors explore some of the security issues, such as cyberattacks and security requirements of V2X. However, some other security challenges related to V2X, such as security concerns

related to infrastructure, databases, and remote keyless entry systems, are not considered. In [21], the authors place a greater emphasis on trust management and do not focus on other related considerations.

The authors of [22] consider the privacy and security aspects of European Union (EU) initiatives across a range of European pilot activities, such as past or present V2X-related EU projects, and analyze ETSI ITS specifications from a security and privacy point of view. However, recent solutions based on cryptography and machine learning are not considered in this study. In [23], the authors survey recent literature on V2X security and provide a review of the most relevant security enhancements. Nevertheless, except for V2X attacks, other security issues, such as infrastructure security challenges and clustering security challenges, are not investigated, and there is no comprehensive description of the recent solutions based on 5G technology. The study in [24] reviews the privacy and security issues for Internet-of-Vehicles (IoV). The authors focus on the effects of new 5G and 6G technological developments for facing IoV security challenges. Indeed, the main focus is on 6G technological enablers for IoV, and the study lacks other security aspects and related countermeasures. The authors of [25] provide a survey on several machine learning approaches applied to networking, communication, and security parts in vehicular networks. Moreover, they review the possible ways of enabling artificial intelligence toward a future 6G vehicular network, including improving network intelligence, enabling the evolution of intelligent radio, and supporting self-learning with proactive exploration. Nevertheless, the main focus is on 6G-enabled mechanisms; other solution techniques, such as cryptography, are not discussed. In [26], the authors provide a systematic literature review for vehicular connectivity and V2X communications. However, this study lacks an investigation of the recent security solutions based on cryptography and machine learning. Moreover, attack classification is not considered. In [27], the authors first introduce the development history of the past Internet of Vehicles, summarize some common V2X security threats, and finally survey the security technologies used for V2X communication. This research lacks a discussion of some other V2X security challenges, such as security concerns related to remote keyless entry systems and databases. There is no attack classification.

Table 1 depicts the investigated surveys in this subsection, focusing on their main contributions and limitations.

C. MAIN CONTRIBUTIONS AND STRUCTURE

This work provides a more focused survey of the vulnerabilities and security challenges in V2X communication technologies. The following are some of the work's essential contributions:

- Providing a comprehensive, well-organized, and detailed overview of V2X communications technology applications, issues, and requirements.

TABLE 1. Comparison of existing surveys: contributions and limitations.

Year	Reference	Contributions	Limitations
2017	[12]	Analysis of most relevant cyber security vulnerabilities, risks, and issues.	The recent solution strategies based on machine learning, 5G and 6G wireless technologies, and blockchain are not considered.
2018	[11]	Attacks description. Investigation of security issues and requirements in cellular networks. Analysis of V2X authentication methods.	The recent developments based on machine learning are not considered.
2018	[14]	Investigation of intrusion detection systems with an analysis and comparison of different detection techniques.	It lacks new recent mechanisms based on applicability of machine learning and cryptography.
2018	[17]	Security analysis related to vehicular networks. Particular focus on anonymous authentication techniques.	It lacks attacks classification and the investigation of other new developments in security solutions.
2019	[15]	Discussion on misbehavior detection mechanisms for insider attacks based on attacker information analysis.	The recent developments based on security solutions for ITS are not considered.
2019	[16]	In depth study of the security features of 5G and their applicability to applicability to V2X communication and vehicular networks.	The recent developments based on machine learning are not considered.
2019	[18]	Comprehensive taxonomy of existing security threats and solutions for V2X. Countermeasures categorization cryptography and identity-based solutions.	It lacks solution methods for attacks against safety critical applications.
2020	[19]	Discussion on security and privacy in V2X. Attacks classification. Countermeasures categorization based on cryptography and trust-based techniques.	It lacks solution methods for attacks against safety critical applications. Moreover, the recent mechanisms based on machine learning and 5G are not considered.
2020	[20]	Security solution classification based on privacy preservation, cryptography, and message authentication. Attacks classification.	It lacks discussion of some other V2X security challenges, such as security concerns related to infrastructure, databases, and remote keyless entry systems.
2020	[25]	Investigation of the possible ways of enabling artificial intelligence toward a future 6G-based vehicular network from the security point of view.	The main focus is on 6G-enabled mechanisms, and there is no discussion of other solution techniques, such as cryptography. Attacks classification is not considered.
2021	[13]	Analysis of security and privacy vulnerabilities in ITS. Discussion of the main challenges and mitigation techniques.	The challenges and mitigation strategies discussed have outdated of modern advancements (machine learning-based strategies, 5G and 6G technologies).
2021	[21]	Investigation of the state of the art solutions on trust management in VANET.	It lacks the analysis of recent trust establishment and management mechanisms. Moreover, it lacks of attacks classification.
2022	[24]	Analysis of privacy and security issues for the Internet of Vehicles (IoV), with a main focus on the effects of new 5G and 6G technological developments.	The main focus is on 6G technological enablers for IoV, and the study lacks other security aspects and related countermeasures.
2023	[22]	A study of the privacy and security aspects of European Union (EU) initiatives across a range of European pilot activities.	The recent solutions based on cryptography and machine learning are not considered.
2023	[23]	A survey of recent literature on V2X security and the most relevant security enhancements.	It lacks the investigation of infrastructure and and remote keyless entry systems' security challenges. There is no comprehensive description of the recent 5G-based solutions.
2024	[26]	A systematic literature review of vehicular connectivity and v2x communications.	It lacks investigation of the recent security solution based on cryptography and machine learning. Moreover, attacks classification is not considered.
2024	[27]	Introducing the development history of the past Internet of Vehicles. Summarize some common V2X security threats. Security technologies used for V2X communication.	It lacks discussion of some other V2X security challenges, such as security concerns related to remote keyless entry systems and databases. There is no attacks classification.

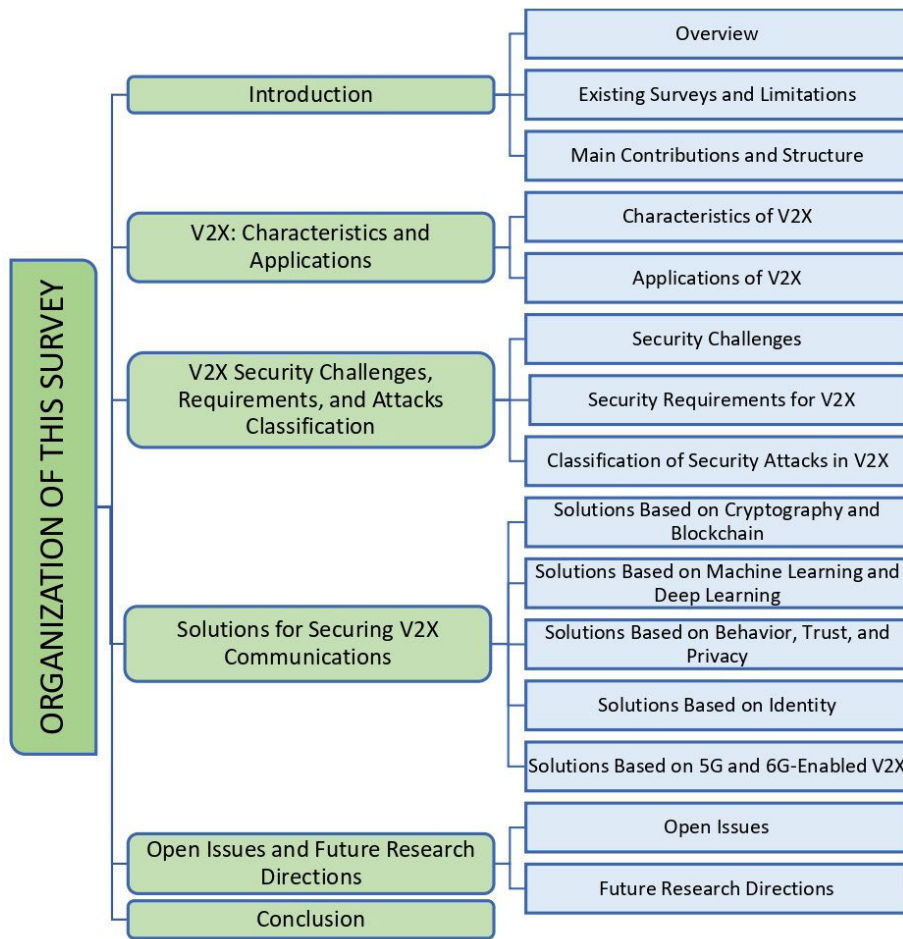


FIGURE 2. The organization of this survey.

- Investigating and classifying the various V2X communication attacks and vulnerabilities as well as the current related countermeasures to mitigate their effects.
- Evaluating state-of-the-art defensive methodologies and solutions. Furthermore, we analyze these methodologies, focusing on their effectiveness and associated limitations.
- Addressing the potential directions for future studies in V2X, particularly in terms of security aspects.

The rest of this paper is structured as follows. Section II describes V2X architecture and characteristics, and some of its applications. Section III presents an extensive V2X security analysis, including the challenges, attacks, and requirements of V2X. A taxonomy of security methods and solutions for V2X technology is presented in Section IV. The research gaps in vehicular communications, as well as perspective future approaches for V2X communications, are discussed in section V. Finally, section VI concludes this survey effort. Figure 2 describes the organization of this research in detail.

II. V2X: CHARACTERISTICS AND APPLICATIONS

The rapidly growing transportation systems are facing several vital issues, including the need to reduce the number of

traffic accidents, rely less on fossil fuels, reduce greenhouse gas emissions caused by pollution from those fuels, improve driver skill levels, and a variety of other closely related concerns. Combined with existing vehicle capabilities, V2X communications have the potential to not only tackle the issues listed above but also to create quite beneficial transportation techniques, improved public transportation services, excellent transportation conditions, appropriate usage of transportation infrastructure, notably precise electronic maps, and several other creative solutions and approaches. In the following, we will organize corresponding (existing) resources in V2X characteristics, and their applications.

A. CHARACTERISTICS OF V2X

Sensing, interacting, and data processing technologies are leveraged by ITS for roadside users, infrastructure units, and vehicles to improve the efficiency and safety of the system [28]. Two major sub-networks constitute the overall network infrastructure providing the needed communication services [29].

- *Intra-vehicle sub-network*: is made up of many sensors that are located in the vehicle. The connection between

sensors is made by Ethernet, ZigBee, WiFi, or Bluetooth connections.

- *Inter-vehicle sub-network*: is responsible for communication between the vehicle and its surrounding devices.

Inter-vehicle sub-network consists of four units (OBUs, RSUs, Road users, and the central/cloud server).

- OBUs: vehicles are equipped with OBUs that allow data processing and interaction with surrounding units.
- RSUs: are pieces of transportation infrastructure located alongside the road, containing data on the local topology that help to provide various services for road users.
- Road users: motorcyclists, bikers, pedestrians, and skaters are examples of road users.
- Central/cloud server: all road users, traffic, and roadways are controlled by the central/cloud server.

The deployment scenarios and traffic characteristics of V2X systems are different from those of other DSRC and cellular network applications [30], [31]. First, the massive density of vehicles, particularly in urban areas, necessitating tiny cells, besides their high speed, results in a dynamical network structure with various mobility patterns. The second thing to remember is that V2X messages are tiny in size and frequently exchanged among vehicles and other V2X communication entities according to an event-driven paradigm or periodically. To enable critical road safety applications as well as other sensitive use cases, like cooperative autonomous driving, the infrastructure must be extremely reliable and provide extremely low end-to-end latency, ensuring timely packet delivery. These requirements are tough to satisfy in V2X because of the short duration of connection time and the high mobility and diversity of the vehicular environment, characterized by frequent contemporaneous vehicle arrival and departure. Briefly, some of the significant characteristics of V2X communication technology are high mobility, dynamic network topology, variable network density, no power limits, critical nature and extensive scalability of the network, extensive computational processing, and vehicle driver protection requirements.

B. APPLICATIONS OF V2X

V2X facilitates the development of a wide range of applications as a consequence of technological advances in wireless sensing and networking. V2X communication applications can be divided into three categories: (i) safety-support applications, (ii) traffic management and road efficiency, (iii) information, comfort, and infotainment services, and (iv) autonomous driving and intelligent traffic systems. These applications are listed in Table 2. For more details, the reader is referred to [32] and [33].

III. V2X SECURITY CHALLENGES, REQUIREMENTS, AND ATTACKS CLASSIFICATION

V2X is extensively utilized in a variety of vehicular communication applications. These applications include traffic safety as well as infotainment applications that impose a diversity of requirements, also in terms of security and

reliability. These various requirements have resulted in the development of many research issues related to V2X challenges. In this section, we investigate the V2X security challenges, requirements, and attack categories.

A. SECURITY CHALLENGES

The security concerns of various types of V2X communications are discussed in this part. The literature on V2X security challenges and countermeasures is presented in Table 3.

1) INFRASTRUCTURE SECURITY CHALLENGES

OBUs interact with RSUs in transferring information regarding road events in a V2I environment. Authenticating OBUs and granting them Internet access are both responsibilities of RSUs [34]. OBUs and RSUs are both subject to malicious activities when operating in a VANET. This section explains how potential solutions can mitigate known vulnerabilities to V2I communications. Several possible V2I attacks are identified by the authors of [35]. The DDoS attack, for example, is defined as transferring redundant information by an attacker to a roadside unit system that prevents it from well-functioning. Some other types of threats are malware attacks, whereby a roadside unit system is infected; impersonation attacks, wherein attackers masquerade as OBUs or RSUs; and eavesdropping attacks, wherein attackers obtain access to private information.

Related Countermeasures: In [35], the authors propose their “CVGuard” enhanced security structure, which, they claim, is helpful to detect and protect against V2I cyberattacks, such as DDoS attacks. The work in [36] presents a software-defined networking-based (SDN-based) detection approach for DDoS threats in V2I communications. Eavesdropping and impersonation attacks can be avoided via authentication mechanisms. In this direction, the Fast Pre-distribution Authentication Protocol (FPAP), which secures interactions between a roadside unit and a vehicle, is introduced in [37]. The authors of [34] introduce a new lightweight V2I authentication system that enables trusted authorities to deliver the group key to OBUs and RSUs in the immediate neighborhood. The authors in [38] discuss an authentication technique based on a dynamic private key that is split in half and handled by a vehicle and an assistance device.

For secured V2X communications, slicing, and network isolation provide more possibilities. Packet loss and overall end-to-end latency may remarkably rise without a network slicing mechanism [39]. For instance, to enhance total bandwidth usage, procedures like flow interaction analysis and network convergence will necessitate isolating and slicing the optical network terminals or units [40]. Moreover, SDNs are able to separate the data planes and control vehicular networks, allowing for such slicing and isolation to be achieved [41].

2) DATABASES SECURITY CHALLENGES

Vehicles that are connected in VANETs face threats and vulnerabilities associated with information transmission to

TABLE 2. Characteristics of V2X applications.

Applications	Channel Models	Benefits	Communication Paradigm	Latency	Packet Loss Rate	Reliability (%)
Road Safety	Wi-Fi, Cellular Network, DSRC, Wave	Collision Avoidance, Incident Management, Sign Notifications	V2V, V2I	Very Low	Very Low	90 - 95
Autonomous Driving	Cellular Network, Wi-Fi, DSRC	Automatic Speed Control and Breaking, Remote Driving, Vehicle Platooning	V2V, V2N, V2I	Very Low	Very Low	90 - 99.99
Traffic Management	Cellular Network, DSRC, ZigBee, WAVE	Traffic Management, Road Monitoring	V2N, V2I	Low	Low	≤ 90
Infotainment, Comfort, Information services	Cellular Network, DSRC, WiMax, WAVE	Route Suggestion, Entertainment, Automated Parking	V2N, V2I	Medium	Low	Not a Concern

cloud-based and on-board databases. While vehicles may communicate data collected from various systems and sensors to their ECUs, they can transmit the information to the cloud in order to be more autonomous and connected. In [42], the authors describe the process as follows: the data collected from sensors is kept on board and analyzed to eliminate redundancies. Then, the data is transferred to edge-based data storage systems or to the cloud for post-processing. Therefore, drivers can obtain service-related information such as personalized routes, parking and weather information, examination of the safety condition of the vehicle, and multimedia content. Together with the standard cloud, the vehicular cloud also exists (a system that combines the resources of a specific vehicle). The vehicular cloud can gather more information than those gathered by a vehicle's diagnostic tools and sensors. GPS data, autonomous driving data, vehicle mobile network data, and sensing data are the four groups into which the authors of [42] divide big data in the field of VANETs. According to the study in [43], the data acquired in VANETs is divided into three categories: vehicle-centric data, which includes data gathered by vehicle sensors and vehicle recorders; user-centric data, which includes information about gender, age, historical data of user access, and education background; and location-centric data, which includes information related to the routes and traffic. DoS, replay, hacking, session key disclosure, and masquerade attacks are some of the attacks targeting cloud-based vehicle databases and are listed in [44].

Related Countermeasures: In [44], the authors propose a technique for authenticating vehicles before the communication of the receiver and sender of the data transmission. Impersonation, replay, manipulation, and masquerading threats are all avoided by this approach. However, the effectiveness of the technique against session key disclosure and DoS attacks was not examined. In [43], the researchers propose a hierarchical database cryptographic system to avoid eavesdropping and tampering threats, and the study in [45] presents an authentication approach that will permit data to be removed if a vehicular system is infiltrated.

3) SECURITY CHALLENGES RELATED TO REMOTE COMMUNICATION TECHNOLOGIES

In V2X communications, remote communication technologies are utilized to transmit information among RSUs and OBUs and to support positioning and localization for

autonomous vehicles. The attackers can use vulnerabilities in these systems to effectively interfere with a vehicle's operation [10]. Some of these remote communication technologies like Remote Keyless Entry (RKE) systems, DSRC/WAVE, cellular networks, Zigbee, Wi-Fi and WiMAX, and Ultra-Wideband (UWB) are all discussed in this section. Indeed, this section focuses on the threats that remote communication systems face and the countermeasures by which they can be protected.

- **Remote Keyless Entry system:** Rather than using a physical key to open a vehicle's doors, RKE systems take the place of that operation. Vehicle doors can be locked or unlocked, engines can be started or stopped, or anti-theft alarms can be activated or deactivated remotely by key holders with RKE capabilities. In RKE systems, accepted button clicks operate a counter employed to generate a rolling code signal that assists in preventing replay attacks [46]. Just a limited number of cryptographic RKE keys are utilized by some companies, like the Volkswagen Group [47]. Rolling code techniques implemented by automakers, like the Volkswagen Group, are targets of eavesdropping attacks and leave automobiles vulnerable to RKE cloning. The author of [48] considers jamming, man-in-the-middle, brute force, and replay attacks as potential risks to RKE. He mentions that some key weaknesses of RKE systems are due to outdated methods and equipment, errors related to implementation, and poor cryptographic algorithms. In [49], the researchers investigate possible assaults on the Hitag2 cipher utilized in various RKE systems, and it has been shown to be vulnerable.

Related Countermeasures: The authors of [50] introduce a Secret Unknown Cipher technique for making RKE controllers clone-resistant. The work in [46] proposes a much more robust receiver that is more resistant to jamming attacks. In [47], the researchers investigate safe cryptographic techniques and secure key distribution. The study in [51] presents a security-conscious RKE approach and a corresponding encryption scheme that resists replay, man-in-the-middle, and brute force attacks. The researcher in [48] also provides a list of solutions for those attacks. In [49], the authors propose a gradual transition to much more secure methods based on the Advanced Encryption Standard (AES).

- **DSRC and Wireless Access in Vehicular Environments:** DSRC is a set of standards that facilitate the transmission of messages with relatively secure and high-speed communications, low latency, and fast network connectivity in V2X communications [52]. The IEEE 802.11p standard, on which WAVE communications are based, guarantees that high-priority communications will not experience delays of more than tens of milliseconds [53]. An attacker may acquire accessibility to the broadcast timetable, which is employed to transmit service notifications and broadcast messages at the same time, resulting in a communication conflict that will go unnoticed by vehicle drivers. In [54], the authors describe the challenges to WAVE service advertisement (WSA). They conclude that WSA is susceptible to attacks on both its accessibility and its security.

Related Countermeasures: In [55], researchers suggest employing a hybrid IEEE 802.11p and Visible Light Communication (VLC) platform to face and overcome IEEE 802.11p weaknesses. Nevertheless, this approach is still vulnerable to replay attacks and packet manipulation. Another hybrid authentication method for RSUs and automobiles with OBUs communication over IEEE 802.11p is proposed in [56]. The authors of [54] examine possible solutions for assaults on WSA availability, reliability, and security. In [52], the authors claim that, as ITS develops to support high-speed and secure communications among roadside infrastructures and moving vehicles, DSRC is not an adequate implementation of V2X communications. As an alternative option, they propose a Heterogeneous Network (Het-Net), which makes use of DSRC, LTE, and Wi-Fi.

- **Cellular:** The bandwidth of DSRC is so limited that it cannot handle upcoming V2X communications [11]. Because of this, specialized cellular infrastructures, such as Long Term Evolution Vehicular (LTE-V), Cellular V2X (C-V2X), and 5G-New Radio (5G-NR) [11] can offer alternate more suitable communications options [53], [57], [58] [11]. The author of [59] addresses security concerns about the lack of cryptographic safeguards in LTE signaling message transmission. There is a comprehensive analysis in [57] on LTE security issues, including eavesdropping attacks, jamming attacks, and attacks launched from exploited femtocells or unlicensed base stations. In [60], researchers investigate the security of LTE from the perspective of encoding users' data and network authentication. Analysis of C-V2X communication challenges is provided in [61]. Both LTE and 5G-based vehicular networks are examined by the authors of [11] in terms of security.

Related Countermeasures: The study in [57] discusses some mitigations and preventive actions to the attacks noticed in LTE communications. The authors

of [62] propose that Software-Defined Networking (SDN) and Network Function Virtualization (NFV) can be used to enhance existing LTE security features. In [63], researchers present a lightweight key management approach to help the authentication of nodes in LTE communications. Cellular communication vulnerability countermeasures are also discussed in works [11], [60], [61].

- **ZigBee:** Zigbee is a short-range communication standard designed for power-saving low-data rate communications in the ISM frequency bands. A ZigBee implementation in Forwarding Collision Warning Systems is presented in [64]. It was originally standardized as IEEE 802.15.1 but has been significantly modified over time essentially for reducing its power demand features. Moreover, due to its universal availability and widespread deployment, Bluetooth is frequently applied in connected vehicles to link smartphones with the vehicle's telematics and infotainment system, allowing calling functionality, automobile diagnostics, calendars, and music streaming [65]. However, Bluetooth is excessively complicated and fragmented and has received insufficient security examination [66]. Generally, Zigbee security is mentioned in relation to the context of the Internet of Things (IoT), and security studies for automotive Zigbee are almost non-existent. The authors of [67] characterize and illustrate three attacks on ZigBee systems. Such threats take advantage of the following weaknesses: first, attackers may acquire devices' configuration information within an in-range ZigBee system; second, several ZigBee systems remain unencrypted; third, the attackers will retransfer ZigBee system traffic to carry out replay attacks. The authors investigate security weaknesses in ZigBee IoT devices in [68].

Related Countermeasures: To tackle the three vulnerabilities they investigated, the authors in [67] recommend using intrusion detection solutions, establishing network keys before installation, and incorporating time-stamping within ZigBee's cryptographic algorithms. In [68], researchers suggest not utilizing pre-loaded and factory-generated keys. As an alternative, they recommend that each ZigBee device be provided with a network security key and static device IDs be replaced with a rotating mechanism of device IDs to improve security.

- **Wi-Fi and Worldwide Interoperability for Microwave Access:** Wi-Fi, standardized according to many IEEE 802.11 variants, could also be used for V2X communications operating in ISM bands or at least as a supplement to DSRC/WAVE. In addition to Wi-Fi, the Worldwide Interoperability for Microwave Access (WiMAX) is a standard that is very similar to but more efficient and faster than Wi-Fi. Security concerns about automotive WiMAX and Wi-Fi have received

scant attention in the literature. Nevertheless, in [69], researchers managed to hack a Tesla Model S remotely in particular by considering the fact that the passcode to an embedded Wi-Fi SSID was stored in plain text. After that, they had the ability to impersonate a hotspot and divert traffic to anywhere they wanted. The authors in [70] investigate the IEEE 802.11 standard, the standard that Wi-Fi is developed on, and discover that the interleaved convolutional technique makes it subject to jamming attempts. In [71], researchers investigate how Evil Twin attacks on Wi-Fi could be prevented. Users are unknowingly linked to an unauthorized Wi-Fi access point through an Evil Twin attack, which may also subsequently spy on their personal activities. DoS attacks on Wi-Fi Protected Access are discussed in [72]. Man-in-the-middle, Radio Frequency Interference, Radio Frequency Jamming, Replay, Eavesdropping, and DoS assaults are some of the challenges to WiMAX security outlined by the authors of [73].

Related Countermeasures: In [70], researchers propose using the random, encrypted combination to resist Wi-Fi jamming attacks. In [71], the authors propose using a different gateway than the actual Wi-Fi access point to detect Evil-Twin attacks. The work in [72] addresses DoS attack solutions. The authors of [73] describe technical countermeasures for WiMAX vulnerabilities and threats through the categories of Client Device Security, Confidentiality and Integrity Protection, Authentication and Authorization, Upgrades, Updates, and Patches.

- **Ultra WideBand:** Another type of wireless communication is Ultra WideBand (UWB), which is famous for its capability to transfer significant information at different data rates on a band more than 7 GHz wide, as well as for its low power consumption [74]. It is recommended for prevention from replay attacks in Passive Entry Passive Start (PEPS) systems, autonomous vehicles' off-road localization, electric vehicle localization, and outdoor localization [74], [75], [76]. There is a deficiency of research on UWB cybersecurity weaknesses. The author of [77] emphasizes the vulnerability of low-power UWB platforms to eavesdropping threats, as low-power devices are unable to sustain the overhead associated with many encryption approaches.

Related Countermeasures: In [77], the author proposes a technique for safeguarding UWB transmitters and receivers against eavesdropping attempts. Rather than depending on cryptographic techniques, UWB communications are secured via time hopping.

B. SECURITY REQUIREMENTS FOR V2X

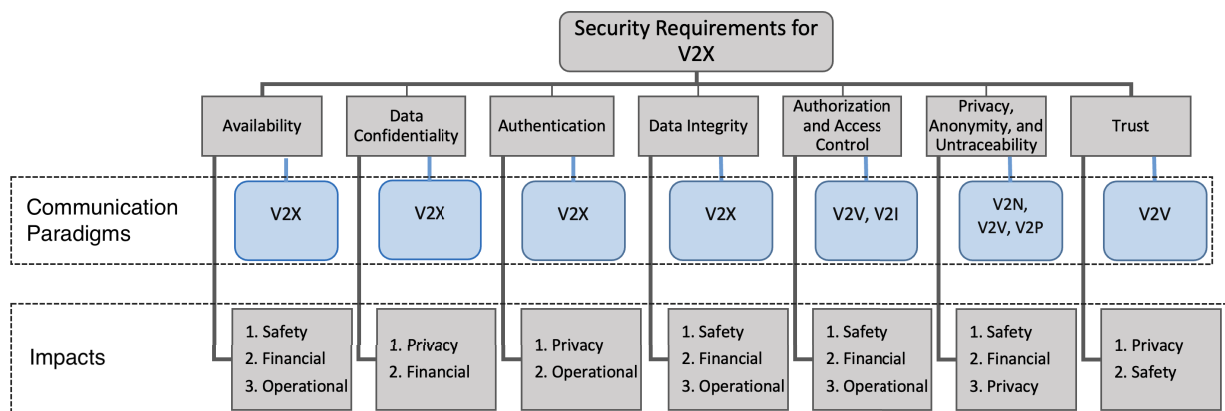
Given the intrinsic broadcast character of wireless systems, which are susceptible to a wide range of network attacks, security is critical in every kind of wireless communication network [78]. Similarly, information sent between V2X objects can be replayed, forged, or eavesdropped on by an

attacker in the V2X environment. Furthermore, attackers may mimic a legal V2X object, such as an automobile, RSU, etc., to transmit incorrect data for re-distribution to other entities of the V2X system. So, the structure of most V2X services necessitates protecting both V2X objects and communications from threats that can lead to fatally incorrect decisions/actions, particularly in safety-critical circumstances. Hence, to protect against various threats, the V2X communication networks should meet the primary security requirements of availability, confidentiality, authenticity, integrity, authorization, and privacy. This will also guarantee the effective implementation and widespread acceptance of V2X technologies and the development of a safer V2X system. Therefore, we provided a brief description of these security aspects in the scope of V2X, as illustrated in Figure 3 and described in [79] and [80].

- *Availability:* This requirement guarantees that communications among V2X objects are analyzed and made available to their intended receivers in a well-timed way. Thus, it necessarily needs the use of lightweight and low-cost cryptography techniques. Additionally, authorized V2X objects must be able to access the system at any time and from any location, including during DoS attacks.
- *Data Confidentiality:* Data transmitted between V2X units must be appropriately encrypted and safeguarded to protect sensitive data from being disclosed to unauthorized units and malicious entities. Nevertheless, because the information transferred in V2X is generally public, confidentiality may not be the highest priority, except for information about the privacy of vehicles' occupants.
- *Authentication:* In V2X communication technologies, authentication is one of the most critical security requirements, as it allows valid objects in V2X to be distinguished from malicious entities while interacting in the V2X communication networks. Authentication may then be subdivided into message authentication and user/source authentication. Message authentication provides the authenticity of message packets so that the recipient is able to check that the received message packet was delivered by the legal units and was not manipulated during transit. In contrast, user authentication establishes the validity of an entity.
- *Data Integrity:* It is necessary for the data transmitted between V2X units to be validated in a timely way to identify any data manipulation, alteration, or deletion during transmission.
- *Authorization and Access Control:* It is required to ascertain that legal V2X elements have access to the appropriate V2X services in accordance with a predefined series of policies and rules. Providing V2X objects access or restricting their access to specific V2X services must be governed by particular rules.
- *Privacy, Anonymity, and Untraceability:* The V2X communications must ensure that network users' privacy

TABLE 3. V2X security challenges and countermeasures exploitation.

Investigated Aspect	Security Challenges	References	Security Countermeasures	References
Infrastructure	DDoS, Impersonation, Eavesdropping, and Malware Attacks	[35]	CVGuard Security Structure, SDN-based Detection Approach, FPAP Protocol, Lightweight V2I Authentication Systems, Slicing and Network Isolation Mechanisms	[35]–[41]
Databases	DoS, Replay, Hacking, Session Key Disclosure, and Masquerade Attacks	[44]	Authentication Techniques, Hierarchical Database Cryptographic Systems	[42]–[45]
Remote Keyless Entry System	Eavesdropping, Jamming, Man-in-the-Middle, Brute Force, and Replay Attacks	[48], [49]	Secret Unknown Cipher Technique, Robust Receiver System, Safe Cryptographic Techniques, Security-Conscious RKE Approach, AES-based Methods	[46], [47], [49]–[51]
DSRC, WAVE	Packet Manipulation, Replay Attack, Security Attacks on WSA	[54]	Hybrid Authentication Methods, Heterogeneous Network (Het-Net) Techniques	[52], [54], [55]
Cellular	Lack of Cryptographic Safeguards, Eavesdropping and Jamming Attacks, Attacks Launched from Exploited Femtocells or Unlicensed Base Stations, Network Authentication Issues	[11], [57], [59]–[61]	Software-Defined Networking-based Techniques, Network Function Virtualization-based Methods, Lightweight Key Management Approaches	[11], [50], [57], [60], [61]
ZigBee	Attacks on Devices' Configuration Information, Attacks on Unencrypted Systems, Reply Attack, Zigbee IoT Devices' Weaknesses	[67], [68]	Intrusion Detection Solutions, Establishing Network Keys before Installation, Cryptographic Algorithms, Pre-Loaded and Factory-Generated Keys	[67], [68]
Wi-Fi, WiMAX	Man-in-the-Middle, Radio Frequency Interference, Radio Frequency Jamming, Replay, Eavesdropping, DoS, and Evil Twin Attacks	[69]–[72]	Using Different Gateways Technique, Random Encrypted Combination Approach, Client Device Security Systems, Confidentiality and Integrity Protection Mechanisms, Authentication and Authorization Techniques, Upgrades and Updates Mechanisms	[70]–[73]
UWB	Reply and Eavesdropping Attacks	[74]–[76]	Time Hopping Technique	[77]

**FIGURE 3. Main security requirements in V2X: communication paradigms and potential impacts.**

is protected. Customer's personal and confidential data and identities must not be revealed. Thus, in a wider perspective, anonymity is regarded as a subclass of privacy in vehicular communication networks, whereas privacy relates to the concealment of information/data. RSUs can trace a vehicle's position and the location of vehicles during the authentication process, infringing on the location's privacy. Privacy solutions must meet the untraceability and unlinkability criteria to maintain privacy. Untraceability implies that a vehicle's activities cannot be tracked, while unlinkability indicates that an unauthorized party should be unable to associate a vehicle's identity with that of its driver or owner.

- **Trust:** Trust is a critical security factor since it helps safeguard the V2X communication system from threats and attackers. Trust can be defined as an entity's confidence in another entity that is a component of the vehicular network. Trust computation is an extra security criterion in V2X communication systems, which is utilized in cooperation with availability and privacy protection to enable extremely protected data transmission. It is critical in vehicular networks to assure truthfulness between vehicles using trust computation.

C. CLASSIFICATION OF SECURITY ATTACKS IN V2X

Regarding the significant importance of security attacks in V2X communications technology, this section represents a

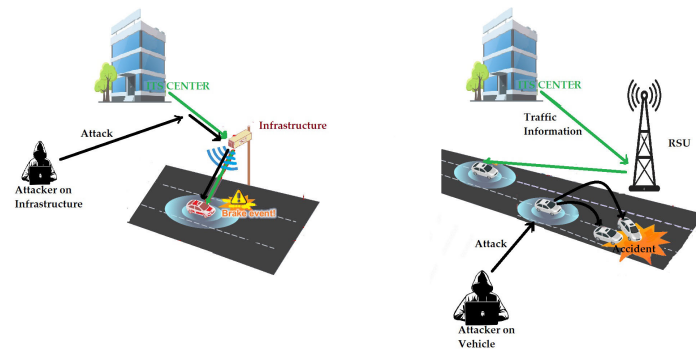


FIGURE 4. Examples of attacks on V2X communications.

categorization of attacks in V2X and vehicular networks based on five categories: attacks on infrastructure, attacks on hardware and software, attacks on privacy, data trust attacks, and attacks based on behavioral patterns [20], [81]. Figure 4 depicts two examples of attacks using V2X communication. In the right picture of Figure 4, the attacker hacks the infrastructure or RSU to send false information to the vehicle or force events, such as hard braking. In the left picture of Figure 4, the attacker's vehicle causes an accident by sending a compromised BSM to nearby vehicles. The literature and a comparative study of the various attacks on V2X communications are presented in Tables 4-9.

1) ATTACKS ON INFRASTRUCTURE

These attacks represent a threat to the infrastructures that are necessary for the proper operation of V2X communication technologies. Hardware tampering, unauthorized access, and other similar attacks are some examples belonging to this category. We list some of them in the following (Table 4).

- *Session hijacking attack*: Typically, authentication is performed at the beginning of each new session. Later on, the attackers take control of the session between the nodes, leading to what is described as a session-hijacking attack on the network. The usage of public key infrastructure and trust authority are two protective strategies against session hijacking attacks. Every node's unique identity is known to the trust authority, which has access to this information. When a vehicle interacts with an RSU, the RSU first checks the vehicle's identification using the trust authority before sharing the key with the vehicle. If the vehicle's identity is valid, RSU will share the key.
- *DDoS attacks*: DDoS attacks are denial-of-service attacks in which a primary entity has the role of attack coordinator, indirectly controlling a large number of other attackers (bots) that behave as a single entity. DDoS attacks, decoupling the real attacker's capabilities from the effects, are primarily responsible for network disruptions since they flood the system with hostile or at least undesired packets. The use of strong authentication strategies and digital signatures at the individual node level can help to reduce the effectiveness of such kinds of attacks.
- *Unauthorized access attack*: It is possible for malicious organizations that do not have the necessary permissions or privileges to access network services through this form of attack. Eventually, this attack leads to property damages, accidents, or the stealing of private information. Confidentiality makes some restrictions on illegal access to private data such as identity, position, and license plate number. The most frequently employed strategy for ensuring confidentiality in vehicular networks is the employment of pseudonyms [82]. While utilizing pseudonyms, each vehicle node contains multiple key pairs with encryption. For data encryption or signing, various pseudonyms are utilized. The vehicle node can't be linked to the pseudonyms, but the relevant authority does have access to them. Vehicles are expected to get an updated pseudonym via RSUs when the previous pseudonym expires so that they can replace it.
- *Hardware tampering attack*: This kind of attack is the result of specific malicious individuals of automotive manufacturers who attempt to manipulate the hardware while doing annual maintenance on the vehicles. The purpose of this attempt is to either obtain access to the vehicle network data or put particular data in the vehicle. Hardware tampering attack effects can be mitigated using the Trusted Platform Module (TPM) [83].
- *Masquerade attack*: In this type of attack, the attacker hides by masquerading or using an authorized identity. It intends to build a Blackhole or create fake data packets that appear to be gotten from legitimate nodes but are fake. Another option is for an attacker to act as a fake emergency automobile and force passing vehicles on the roadway to change directions or slow down their speeds. The implementation of non-repudiation mechanisms in V2X communication can defend against masquerade attacks.

TABLE 4. The literature and a comparative study of security attacks on V2X communications (Attacks on infrastructure).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Session Hijacking	Infrastructure	Privacy, Availability	Taking control of session	High	[20]
DDoS	Wireless Interface	Availability, Authentication	An attacker executes the role of attack manager along with other attackers	High	[20], [36], [84]
Unauthorized Access	Infrastructure	Privacy, Confidentiality	Forceful access of network services	High	[82]
Hardware Tampering	Hardware and Software	Privacy, Confidentiality	Gaining access or placing specific information from/to vehicle	High	[83]
Masquerade	Hardware and Software	Integrity, Authentication, Non-Repudiation	Using a mask or valid identity for hiding itself	High	[20]

2) ATTACKS ON SOFTWARE AND HARDWARE

This section discusses potentially harmful attacks on the software and hardware of the vehicle's various control systems (Table 5).

- *DoS attacks*: DoS attacks are a kind of attack that aims to disrupt the availability of network services. These attacks have the potential to have a detrimental effect on the performance of applications in automotive networks. The attackers can be external or internal. Their principal purpose is to disrupt communication tools and routine services, making them unavailable to authenticate users. The flooding attack, in which the attacker overflows the control channel with a massive quantity of messages, is an example of a DoS attack. As a result, RSUs and OBUs are unable to process such a large number of data, causing network disruptions. Digital signatures and the usage of particular authentication mechanisms can help to mitigate DoS attacks. Tesla++ is an instance of authentication being employed as a protective strategy against a DoS attack, where it employs symmetric encryption with delayed key disclosures. In [85], the authors develop a strategy for defending against DoS attacks by utilizing short-lifetime private and public key combinations with a hash function.
- *Spoofing and forgery attack*: In a spoofing attack, the attacker gives the vehicles in the network erroneous position information. It is important to remember that, in vehicular communications, the vehicle's location information must be precise. In these environments, incorrect information concerning vehicle position can result in harmful behaviors. Spoofing attacks can make it simpler to conduct other attacks that employ vehicle authentication as an attack launch platform. Using sign warning messages, utilizing vehicular public key infrastructure for vehicles' communications, constructing group communications, together with a non-cryptographic checksum for each sent data packet, and implementing plausibility checks on the incoming data packet are all possible strategies for defending against message spoofing attacks [86]. Moreover, for supporting the vehicle in determining the exact location of malicious vehicles, further possibilities

include the use of onboard radar or cryptographic certificates [86], [87].

- *Man-in-the-Middle (MiM) attack*: There are many different ways this attack could affect vehicular network environments. The attacker inserts himself between an interacting pair of nodes, in other words, between the receiver and the sender of packets. Furthermore, the attacker can take control of the interaction between two nodes that are interacting with each other. The MiM attack severely affects vehicular systems' authenticity, non-repudiation, and integrity. The effects of MiM attacks are mitigated by employing mechanisms such as strong authentication and secure communication utilizing sophisticated cryptography [88]. A new collaborative message authentication system is described in [89], which allows users of vehicles to cooperatively authenticate a bunch of packet signatures.
- *Brute force attack*: Even though this form of attack is difficult to carry out in vehicular networks because of short connection time and resource limitations, it can nevertheless have an impact in certain conditions. For instance, while attempting to violate the network identification of a vehicle using a specific searching procedure, a brute force attack may be launched. It is possible that executing brute force attacks will compromise the privacy of communications and authentication strategies. To mitigate brute force attack effects, key generation techniques and robust encryption mechanisms that cannot be cracked in a reasonable period should be utilized. These strategies prohibit unauthorized access to the system and data [90].

3) ATTACKS ON PRIVACY

These attacks are a violation of the privacy of vehicular networks' users and the drivers of vehicles that are targeted by these attacks. As explained below, position-tracking attacks and identity-revealing attacks are two instances of privacy attacks that can occur in the V2X environment (Table 6).

- *Identity revealing attack*: During this attack, the identity data of the vehicle occupants is put at threat. The private data of the vehicle occupants has been compromised, and this could have dire ramifications in the future

TABLE 5. The literature and a comparative study of security attacks on V2X communications (Attacks on software and hardware).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Dos	Wireless interface	Availability, Authentication	Targeting network service availability	High	[85]
Spoofing and Forgery	Hardware and Software	Integrity, Authentication, Availability	Providing incorrect location information	Moderate	[86], [87]
Session Middle	Man-in-the-Wireless interface	Integrity, Authentication, Non-Repudiation, Confidentiality	Taking control of the communication between two communicating vehicles	High	[88], [89]
Brute Force	Wireless interface	Confidentiality, Authentication	Effecting the network services	Moderate	[90]

TABLE 6. The literature and a comparative study of security attacks on V2X communications (Attacks on privacy).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Identity Revealing	Wireless interface	Privacy, Authentication	Compromising the personal information of the owner	High	[20], [91]
Location Tracking	Wireless interface	Privacy	Tracking the location and path followed by the vehicle	High	[20], [59], [92]

for them. The usage of an authentication system that includes privacy-preserving techniques can mitigate the identity-revealing attack effects.

- *Location tracking attack*: In this attack, the attacker tracks the position of a vehicle as well as the route passed by the vehicle in a period. By employing temporary and anonymous keys, the user's information is protected from unauthorized access. Consequently, the vehicle's position privacy is preserved, and malicious nodes are unable to track the vehicle's movements.

4) DATA TRUST ATTACKS

The data in transit is altered or modified due to these attacks. As a result, the integrity of data packets is jeopardized. This type of attack can take place in all scenarios of V2X communications, but it is more common in V2I communications. Message tampering attacks, masquerading attacks, hidden vehicle attacks, replay attacks, and illusion attacks are the main types of data trust attacks in vehicular systems. Some of them are discussed in the following (Table 7).

- *Message tampering attack*: This attack leads to the alteration, deletion, modification, and construction of already-existing data. This attack is launched by the attacker altering or rebuilding a particular component of the message packet in order to carry out its harmful objectives. Data correlation method and challenge-response authentication approaches are used to combat message tampering attacks [93]. The data correlation technique described in [93] is based on a security paradigm for implementing a novel group signature scheme. To identify tampered messages generated by unauthorized nodes, this framework makes use of authenticity, anonymity, accountability, integrity, probabilistic signature verification, and access control mechanisms.
- *Hidden vehicle attack*: False location alerts are produced by hidden automobiles in this form of attack, resulting

in accidents. The Global positioning system (GPS) does not perform well, so attackers cheat with GPS data, which is also called GPS spoofing [81]. The implementation of a trust management strategy makes it possible to implement a defensive system against hidden vehicle attacks [94].

- *Illusion attack*: This attack can potentially compromise data trust and the integrity of vehicular communication systems. In this case, the attackers produce incorrect and fake information. Such information provided by the attackers, once spread over the network, relies on communication among vehicles to result in wrong decisions. Since the attacker connects to the system in a legitimate manner, it is challenging to identify it. The illusion attack is avoided by a positioning system with a signature that only accepts authorized position information. Furthermore, differential tracking can be used to detect irregular alterations in the location [95].

5) ATTACKS BASED ON BEHAVIORAL PATTERNS

These kinds of attacks affect the users based on their behavior patterns. The purpose of these attacks is to influence the behavior of the users so that they can launch attacks. There are two different kinds of behavioral pattern-based attacks: malicious attacks and selfish attacks.

a: SELFISH ATTACKS

Such attacks cause nodes to behave selfishly, such as not transmitting packets or not performing verification tasks. The following are some types of selfish attacks that are common in V2X, as well as possible countermeasures (Table 8).

- *Message spoofing attack*: It has already been investigated earlier, along with existing countermeasures.
- *Traffic analysis attack*: In the vehicular context, mobility monitoring or traffic analysis attacks pose a hazard to the users' privacy and also the security of messages that are transmitted. It is an inactive kind of attack in which the attacker monitors a network and, after that, analyzes the

TABLE 7. The literature and a comparative study of security attacks on V2X communications (Data trust attacks).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Message Tampering	Hardware and Software	Integrity, Authentication, Repudiation, Availability	Modifying, altering, deleting or constructing the already existing data	High	[93]
Hidden Vehicle	Hardware and Software	Integrity, Authentication, Repudiation	Generating false position of the vehicle	Moderate	[81], [94]
Illusion	Sensor input in vehicle	Integrity, Authentication	Generating false data	Moderate	[95]

information gathered while monitoring. The findings of the analysis and processing are then used by the attackers to launch subsequent attacks. The majority of studies for resisting traffic analysis attacks focus on preserving privacy, which is achieved through the use of anonymous key configurations that are dynamic and vary based on vehicle speed or through the use of pseudonyms or group signatures [93], [96].

- *Eavesdropping attack*: This is an inactive type of attack in which the attacker listens to and observes the communication channel without informing the target of the attack. This attack compromises the secrecy of transmitted data. This attack enables the acquisition of specific user data that could be used to assist vehicle monitoring. Eavesdropping attacks can be reduced by implementing the previously discussed privacy-preserving procedures and also by encrypting the information. The authors of [97] suggest an asymmetric cryptography solution based on the employment of a non-disclosure routing algorithm, but the authors of [96] recommend using symmetric encryption for beacons to prevent tracking.
- *Repudiation attack*: Tracking events are lost due to a repudiation attack, where a network node refuses to communicate with other nodes. The authors of [98] suggest using trusted hardware, which ensures that any changes to current values and procedures are only made by authorized users to defend against repudiation attacks.

b: MALICIOUS ATTACKS

In this case, the attackers carry out malicious actions in the system, such as message manipulation and replaying. Malicious attacks in V2X are discussed below, along with their countermeasures (Table 9).

- *DoS attack*: It has previously been covered in detail above, along with appropriate countermeasures.
- *Message replay attack*: This type of attack is pretty ubiquitous in all kinds of networks. Indeed, the attacker replays a previously delivered data packet at periodic intervals. An example would be the attacker replaying beacon packets to track down the vehicles' position. Using timestamps on critical messages and timestamping all packets by using transmission time protects the network against replay attacks. A further method is to digitally sign and include the sequence number in each communicating packet [86].

- *Sybil attack*: The attacker creates many vehicles on the roadway with similar Identifiers and, as a result, other vehicles on the roadway are deceived, and data packets are sent to wrong and bogus recipients, ending in the attacker's profit. Several strategies for combating Sybil attacks have been suggested in the literature. The authors in [99] demonstrate how the position of vehicles is utilized to mitigate Sybil attacks by verifying the vehicles' logical positions. When a vehicle receives a message, it assesses the message's certificate, location, and lifetime. If the received packet is in a rational position and is accurate, the recipient accepts it. If not, the recipient notifies the closest certification authority of the unacceptable event.
- *Malicious code attack*: The malicious vehicles, in this sort of attack, transfer malicious data in the shape of codes, like worms, Trojan horses, spyware, and viruses, to assault the base stations and vehicular network. Moreover, malicious codes are capable of destroying vehicle applications and causing vehicle services to be disrupted. This attack can also help in collecting crucial data about trusted vehicles in V2X platforms. A feasible strategy to resist malicious code attacks in V2X systems is implementing a strong security and privacy preservation system in networks providing advanced malware detection capabilities, which can be implemented in a variety of ways.
- *Blackhole attack*: In blackhole attacks, the attacker collects messages from networks but does not participate in the transmission of the data that is received. The routing reports are not updated periodically and reasonably due to this problem. Consequently, authorized users are often prohibited from receiving critical data since the attacker identifies itself to other network parts as a component of the network, but in reality, it is not. Hybrid intrusion detection mechanisms and secure routing topologies are used to counter blackhole attacks in V2X scenarios.

IV. SOLUTIONS FOR SECURING V2X COMMUNICATIONS

Indeed, vehicular communications face numerous security obstacles and challenges. Consequently, guaranteeing communication security in vehicular networks is a complicated task. Due to a deficiency of security solutions in all V2X-enabled scenarios, some additional security strategies are required to safeguard vehicular communication networks. Some of the different security solutions suggested for

TABLE 8. The literature and a comparative study of security attacks on V2X communications (Selfish attacks).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Message spoofing	Hardware and Software	Integrity, Authentication, Availability	Providing incorrect location information	Moderate	[86], [87]
Traffic Analysis	Wireless interface	Privacy, Confidentiality	Listening over a network and analyzing the data collected during the listening period	Low	[93], [96]
Eavesdropping	Wireless interface	Privacy, Authentication	Listening to the communication medium for gathering useful information	High	[96], [97]
Repudiation	Wireless interface	Non-Repudiation	Resulting in loss of event traceability	Moderate	[98]

TABLE 9. The literature and a comparative study of security attacks on V2X communications (Malicious attacks).

Attack	Threat to	Compromised services	Role of Adversaries	Attack Severity	References
Dos	Wireless interface	Availability, Authentication	Targeting network service availability	High	[85]
Message Reply	Hardware and Software	Integrity, Authentication, Non-Repudiation	Resending previous messages at regular intervals	Moderate	[86]
Sybil	Wireless interface	Availability, Authentication	Creating several vehicles with the similar identity	High	[99]
Malicious Code	Wireless interface	Availability	Transmitting malicious data in the form of codes	High	[20]
Black Hole	Wireless interface	Availability	Denying the routing of the received packets	Moderate	[20]

vehicular communication networks are discussed in this section. We classified them into five types based on their security mechanism: solutions based on cryptography and blockchain, solutions based on machine learning, behavior-based solutions, identity-based solutions, and solutions based on 5G and 6G-enabled V2X (see Table 10).

A. SOLUTIONS BASED ON CRYPTOGRAPHY AND BLOCKCHAIN

By designing methods that prohibit unauthorized entities from accessing the system, cryptography is capable of ensuring secure communications between sender and recipient. As a result, such solutions are focused on defending against external threats made by unauthorized entities. Encryption is the principal tool used in cryptography-based methods. It employs a variety of protocols to convert data into a format that is only readable by authorized recipients. For instance, authors in [100] suggest a lightweight, secure navigating model for VANETs. Additionally, the suggested system in [101] makes use of enhanced encryption standards to ensure the security and privacy of users. Authors of [102] present a strategy for achieving security in a vehicular network by utilizing multiple operational channels for the transmission of encrypted information.

The cryptographic keys of a cryptographic system are managed through the key management procedure, which involves key producing, exchanging, and storing. In [103], the researcher presents a novel approach for creating cryptographic mix-zones resistant to malicious eavesdroppers. Furthermore, authors in [104] present a novel key management strategy for the exchange of keys across security managers operating in heterogeneous systems. They utilized blockchain technology to enable secure key exchange within the security

management system. The authors of [105] describe a key generation mechanism that relies on received signal strength to ensure randomness in key generation. Furthermore, the research in [106] describes a system that enables vehicles to create a shared secret key using exchanged signal strength values with a lower chance of obtaining the identical key by surrounding vehicles.

Several authentication approaches that employ a variety of methods to detect and prohibit unauthorized nodes from conducting malicious attacks have been presented. The authors of [107] present two anonymous authentication mechanisms for the V2X communication networks, which are secure but straightforward. Moreover, in [108], researchers suggest a lightweight and anonymous authentication methodology relying on smart card technology. It is divided into two different stages: information authentication and user authentication. The authors of [109] develop a model that meets the privacy criteria. The pseudonym mechanism and identity-based signature are the two mechanisms that make up this model. Additionally, it secures in-vehicle interactions.

The authors of [110] propose a blockchain-based strategy to make V2V communications more secure. In this strategy, all vehicles transmit their position data using beacon messages, and a digital proof, which is a position certificate, is created. This strategy consists of two phases. In the broadcasting phase, a vehicle transmits event messages to surrounding vehicles when an event occurs. Some of the data features like the type of event, a trust level, proof of location, and a pseudo-ID are contained in the event message. By analyzing the event message in the phase of mining, the neighboring vehicles can determine the trust level of the transmitter vehicle. These vehicles make use of message verification rules to determine the trustworthiness of the

TABLE 10. Summing-up of research works on security solutions for V2X (Categorized based on the solutions).

Solution based on	Employed Technique	References
Cryptography and Blockchain	Encryption	[100]–[102]
	Key Management	[103]–[106]
	Authentication	[86], [89], [107]–[109]
	Blockchain	[105], [110]–[116]
Machine Learning and Deep Learning	Supervised Learning	[117]–[126]
	Unsupervised Learning	[127]–[130]
	Reinforcement Learning	[84], [93], [114], [131]–[133]
Behavior, Trust, and Privacy	Weighted-Sum	[134]–[139]
	Rewarding Mechanism	[140], [141]
	Privacy Preserving	[142]–[146]
	Fuzzy Logic	[147]
Identity	Geographic Proximity	[148]
	Random Pseudonyms	[91], [92], [149]
	Group ID	[150]
5G and 6G Technologies	5G-enabled V2X	[151]–[155]
	6G-enabled V2X	[156]–[158]

message. The authors of [111] introduce a blockchain-based method for securing communications and payments and investigate the usage of blockchain as middleware between various participants in ITS. Another structure, which is based on blockchain for enhancing the security of information transmission in V2X communication networks, was given in [112]. According to this work, typical security defensive methods are weak and unsuccessful primarily due to their centralized nature, in which cloud servers act as a bottleneck.

In [113], the authors propose a trust management mechanism for vehicular ad hoc networks that is based on blockchain. In this mechanism, the vehicles' trust values are stored in a blockchain and shared among RSUs. In [115], the authors investigate the relations of IoT, V2X communications, and blockchain technologies and discuss several V2X security issues and their countermeasures. Then, various applications of V2X and their related services are described. A focused survey on blockchain and SDN-based vehicular networks, analyzing control-plane architectures, security issues, and specific challenges has been presented in [116].

B. SOLUTIONS BASED ON MACHINE LEARNING AND DEEP LEARNING

Machine Learning (ML) has become a popular and practical strategy for providing effective solutions across a wide variety of application areas. Vehicular networks and V2X communications are essential application areas where ML-based strategies have proven to be quite beneficial in solving a variety of challenges. Nowadays, there is a growing interest in applying ML techniques to vehicular network security to obtain better, faster, and more accurate attack detections and predictions.

As mentioned before, a significant task in the vehicular system is detecting and eliminating anomalies during data transmission between various systems or components. Some of the most common ways to detect anomalies in a data stream are through 1) rule-based learning, which validates each assessment based on existing knowledge, 2) cross-checking the metrics (for example, speed and Location) among various data streams from sensors, or 3) observing the data stream across a sliding window of time in order to spot abnormal trends. Due to the volume and variety of information, it is frequently challenging to construct handwritten rules to identify aberrant signals in each occurrence of a data stream. A comprehensive survey on ML-based Misbehavior Detection Systems has been presented in [159]. It classifies state-of-the-art solutions by discussing and analyzing them from both security and machine learning point of view.

A pragmatic application of ML strategies is to substitute human-written rule-based invasion detection strategies. This application will use ML and Deep Learning (DL) to discover features in massive sample datasets used for training. The employment of feed-forward and recurrent DL techniques for invasion detection has been described in recent papers such as [117] and [118]. In comparison to rule-based approaches, strategies based on ML and DL methods will provide many benefits. Using the same detection methods for different vehicular communication systems is one of these advantages. An ML-based detection system is data-driven, which means that it creates rules based on the data. Applying ML techniques can make developing platform-specific security software for any vehicle sub-system simpler.

Another advantage of employing ML models is robust detection. Deep learning models comprising convolutional neural networks (CNN) and long short-term memory (LSTM)

layers can be applied to sophisticated architectures that conduct robust detection by utilizing data from various temporally and geographically distinct data streams. ML and its different architectures are growing in popularity in V2X communication networks for detecting threats and facing various kinds of security challenges. The work in [119] describes a data-centric method for detecting misbehavior for the internet of vehicles. In [127], the authors present an unsupervised model for detecting radio frequency jamming in V2X communications. Authors of [128] suggest a methodology based on ML for detecting four distinct kinds of attacks on a vehicle's cruise control mechanism (including Velocity, Acceleration, Position, and Velocity-Position). The work [129] presents a data-driven strategy for detecting DoS and three different kinds of in-vehicle network attacks. In this work, unsupervised learning is used to extract relevant features from information correlated with the controller area network (CAN) bus. The authors of [84] introduce a method for detecting vulnerabilities by examining live network traffic packets.

In [114], the authors consider the provider's concern about privacy and propose a blockchain-based federated learning model for secure information transmission. The authors of [120] suggest using collaborative learning, wherein vehicles exchange their experiences with others, to guarantee improvement in malicious vehicle detection. Cooperation between automobiles also raises privacy concerns.

The work [160], introduces a data-oriented trust mechanism to assist in safeguarding decision-making units from fake data. In this mechanism, a vehicle asks from the trust evaluation model for trust values related to a driving event like speed regulation and path selection. The authors of [121] utilize fuzzy logic and an ML model to do trust computation. They suggest a trust computation technique based on fuzzy logic to access the accuracy and integrity of event messages and their transmitters. The authors of [131] propose a Q-learning approach to evaluate the trustworthiness of automated driving vehicles and identify invaders according to the calculated level of confidence. This research addresses two distinct assessment methodologies, referred to as indirect and direct automated driving vehicle examination methods.

Simulating attacks to evaluate detection systems is just as important as developing new systems. This simulation necessitates a relatively efficient mechanism for creating high-quality synthetic incursion signals. Generative systems, like Generative Adversarial Networks (GANs), can be employed to create statistically identical invasion signals. Comprehension, perception, and projection of an environment's dynamics are referred to as situational awareness. In the case of vehicular networks' security, situational awareness can refer to a security strategy in the vehicular cloud that knows the VANET architecture and communication channels and is also critical for threat analysis and mitigation. This issue is much more challenging and complicated than intrusion detection because the strategy must make predictions in a constantly changing environment with unreliable and

dynamic data. However, these applications can benefit from a variety of ML models. In [132], the authors present an enhanced and robust physical authentication system based on RL to safeguard against unidentified spoofing attacks in vehicular networks. The study in [133] considers the optimal routing selection problem in software-defined vehicular networks, where RL is utilized to determine the most secure communication link policy under the effect of malicious vehicles.

Countermeasures for situational awareness and network security technologies are, for their nature, defensive methods, and even the finest defensive strategies are susceptible to failure. Thus, an alternative and secondary solution is required to prevent the vehicular system from making irrational actions that imperil its occupants and the users of other vehicular systems. Suppose a GPS spoofing attack is successful, and the automobile is supplied coordinates that cause it to drive into the boundary. A fallback mechanism can withstand this attack, as the vehicular system will recognize safe, suitable, and dangerous drivable parts of the roadway. Therefore, the vehicle should have some perception and knowledge of its surrounding environment and road conditions. Several CNNs have been developed mainly to segment driving environments in real-time and with acceptable accuracy [122]. One of the other perceptive challenges is to forecast the speeds and paths of surrounding vehicles to evaluate their dynamics. Previous studies, such as [123], employed probability-based ML to estimate vehicle movements and paths, while more recent studies, such as [124], use recurrent deep neural network models. In the context of the vehicular cloud, perception can take the shape of traffic flow forecasts, which is a topic in which new studies have made use of DL techniques [125].

In V2X communication networks, to avoid hacking and vehicle stealing, autonomous vehicles must retain the true identity and accurate profiling of the user/driver. Scientists are using ML approaches to validate the true identity and fingerprinting of a user/driver to make vehicles resistant to stealing while hiding the correct identity of users/drivers. In [130], to identify and classify drivers, deceleration, and acceleration information are analyzed. Indeed, to identify drivers, the authors suggest a multi-class linear discriminant analysis classifier. The authors of [126] conduct an analytical investigation of ten alternative ML models to identify the true user/driver efficiently. The goal of this research is to utilize the possible minimum number of features to predict driver behavior, which contains friction torque and intake air pressure.

C. SOLUTIONS BASED ON BEHAVIOR, TRUST, AND PRIVACY

As a complement to previous solutions (particularly cryptography), behavior-based and trust-based approaches are recommended. Typically, trust assessment is carried out by monitoring nodes, which continuously observe and aggregate information about other nodes' activity and behavior information. In the following, the most often utilized approaches

for vehicular network trust management will be discussed. The weighted-sum methodology is the most frequently used strategy for trust management, in which trust is evaluated by allocating various weights to any trust component. Whenever a node engages in malicious behavior, the overall trust value reduces until it becomes the minimum value or, in other words, zero.

The authors of [134] introduce a trust system that presents two trust metrics known as RSU-based and vehicle-based. Every node can observe and check its surroundings and evaluate the value of local trust. The work in [135] employs an ant colony optimization approach to isolate non-cooperative nodes and find the shortest trusted route. Several models have been introduced for monitoring traffic events and evaluating their trustiness in vehicular networks. In the presence of inconsistencies, the security mechanism is utilized to find the malicious node. Moreover, a similarity mining method is proposed in [136] to discover similarities between vehicles and message packets. As an example, messages regarding the same event produced by the same automobile, in the same place, and at the same time usually have equal trust values.

In [137], the authors introduce a mechanism for determining the trusted area utilizing the Ad-Hoc on Demand Distance Vector routing algorithm, whereas RSUs are in charge of monitoring packets' routing. The vehicles may, however, take the wrong route as the population of malicious nodes rises. The structure suggested in [138] is made up of two segments. The first one performs three security checks to evaluate the trustworthiness of the message. Firstly, it verifies that the communicated message packet originated from a trusted source and is transmitted through a trusted path. Secondly, it checks whether any malicious nodes are present along the message packet route. Thirdly, it examines to see if the packet content is legitimate. In other words, it calculates a trust value for each neighborhood and each road part. Whenever a message packet has been analyzed and determined to be trustworthy, the system searches for a trusted path to transmit the message packet.

In order to escape from detection, the most recent and advanced malicious nodes exhibit sophisticated behavior and hide themselves by intermittently changing between malicious and normal behaviors. Trust value, distance, and load are three main factors that the model takes into account while selecting potential verifiers. Authors of [139] present a distributed authentication strategy in which verifier nodes examine the message's validity, and non-verifier nodes rely on the verification findings. Several techniques are used to select verifier nodes, including the most-even distributed, N-nearest, and hybrid ones.

Credit is used to reward cooperating nodes in the rewarding-based strategies. In other words, the node rewards its nearby nodes by behaving normally and cooperating with other nodes. Consequently, a node that has a high trust value is regarded as a trustworthy and reliable node. As a security mechanism, the rewarding technique persuades non-cooperative nodes to engage in the packet transmission

process. The authors in [140] develop a Payment Punishment System used during the selection, routing, and transmission processes. The study in [141] provides a trust-based routing algorithm with two significant stages. The trust measurement is conducted frequently and in a distributed way in the first stage. Then, based on the result of the previous stage, the second stage includes transferring the data packet over the most secure and trusted path.

Instead of attempting to design a system mathematically, the fuzzy logic methodology uses a set of fuzzy rules to resolve management, control, and security problems. In the models based on fuzzy logic, the fuzzy sets, requirements, and criteria are established first. After that, the initialization of the input variables' values is done. Finally, the fuzzy system characterizes the output data and assesses the results by employing the fuzzy rules. Fuzzy logic models for detecting packet-dropping assaults are described in [147].

As mentioned, security, privacy, and trust are three fundamental concerns in any vehicular network application that must be addressed. The essential security issue is ensuring the security of V2X communication channels. Privacy implies that only authorized individuals within VANETs can possess the authority to access and control vehicle data, including actual identification and location information. In VANETs, trust management refers to the mechanisms by which a vehicle assesses the trustworthiness of other vehicles and the communications it receives. In the survey paper [142], the authors investigate these security aspects in vehicular networks. Indeed, various location privacy safeguarding strategies utilizing pseudonymity are reviewed to enhance vehicle privacy while concurrently ensuring the quality of location-based services. Furthermore, the authors provide an exhaustive assessment of several trust management mechanisms in VANETs. In conclusion, given that existing and future applications in VANETs are assessed by simulation, the authors provide a timely update on the most recent mobility and network simulators and the integrated simulation platforms.

The authors of the article [143] introduce an innovative privacy-preserving trust management (PPTM) strategy for the transmission of emergency messages in space-air-ground-integrated vehicular networks (SAGIVNs). The suggested method provides accurate trust management and practical conditional privacy preservation concurrently with decreasing communication overhead while also offering significant applicability, strong robustness, and several other appealing attributes. Additionally, the comprehensive theoretical analysis and simulation assessment are elaborated upon. The findings demonstrate that the suggested method outperforms the current techniques. In [144], an innovative method called BTMPP (balancing trust management and privacy preservation) is proposed, which effectively delivers both accurate trust management and robust conditional privacy preservation for emergency message dissemination in vehicular networks. This is achieved by utilizing the commonly used bloom filter-based private set intersection technology, which

effectively balances privacy preservation and trust management in vehicular networks. The research [145] introduces a lightweight privacy preservation strategy with effective reputation management (PPRM) for mobile crowd-sensing in vehicular networks. It provides a lightweight technique for privacy-preserving sensing task matching that safeguards identity privacy, location privacy, reputation value privacy, and sensing data privacy while decreasing computational and communication overheads for sensing vehicles. Specifically, to prevent the forgery of reputation values and to identify trustworthy sensing vehicles, the authors propose a privacy-preserving method for verifying the equality of reputation values and a privacy-preserving algorithm for proving the range of reputation values for choosing reliable sensing vehicles. A three-factor reputation value updating technique is then developed to effectively and precisely update the reputation values for sensing vehicles. In the research paper [146], the authors present a privacy-preserving reputation updating (PPRU) approach for cloud-assisted vehicular networks utilizing Elliptic Curve Cryptography and Paillier techniques. The reputation feedback is gathered, handled, and controlled by the honest but curious Cloud Service Provider in a privacy-preserving way, significantly decreasing the communication and computational overheads on the Trusted Authority side. Simultaneously, the suggested method offers robust privacy protection, strong security, and effective reputation management with relatively low computational and communicational overhead.

D. SOLUTIONS BASED ON IDENTITY

They tackle attacks that utilize users' identities for harmful purposes, like breaching users' privacy and launching Sybil attacks. In these attacks, an attacker gains access to users' privacy and exploits it. The security mechanism should function in an anonymous identity environment to deal with this situation. Most security countermeasures use the vehicle's identity to detect these kinds of attacks and discard malicious nodes.

In [149], the authors provide a trust management strategy that considers anonymous identity. By using this strategy, vehicular ad hoc networks are protected against transmitting messages with phony times and locations. In [150], the authors suggest a privacy-preserving trust mechanism that provides data collection on vehicle behaviors while protecting privacy and confidentiality by using a group ID instead of an actual identity. In [92], the authors present a protective strategy for some types of eavesdropping attacks. In a specific scenario, an attacker traces a target vehicle using a particular pseudonym.

The authors of [91] introduce a security mechanism for VANETs based on three basic strategies. In the first one, a pseudonym-based strategy is employed to provide private/pseudonym key pairs to vehicles traveling within or beyond domain boundaries. In the second one, the threshold signature strategy is used to transmit the confidential information necessary for recovering the identity of a malicious

vehicle. Furthermore, it prohibits compromised nodes from compromising other ones. In the third one, a defensive strategy based on threshold authentication enables a system to distinguish between malicious behavior and malfunctioning. Whenever a malicious node utilizes several identities, the geographic proximity strategy is utilized to detect Sybil attacks. This strategy is used by authors of [148] to detect Sybil identities when the infiltrated node's geographic proximity and most of its Sybil identities remain over a long period and are repeated.

E. SOLUTIONS BASED ON 5G AND 6G-ENABLED V2X

Recently, communication technologies have grown at an incredible rate, significantly improving the capabilities of the intelligent transportation system. These developments are expected to continue, particularly as the Fifth and Sixth Generation Wireless Technologies (5G, 6G) are actively developed and deployed [161], [162]. The 5G and 6G architectures incorporate some new security components: Authentication Server Function (AUSF), Security Context Management Function (SCMF), Authentication Credential Repository (ARPF), Policy Control Function (PCF), and Security Anchor Function (SEAF). The feasibility of integrating V2X communication protocols with 5G appears promising due to the 5G Infrastructure Public Private Partnership (5G PPP). It is interpreted as a tremendous opportunity to develop more innovative, efficient, and flexible services to facilitate the move to higher degrees of automation while maintaining the sustainability, efficiency, and safety of transportation systems [151], [152].

Nowadays, the link-layer protocol utilized in V2X communication is 802.11p, which covers traditional techniques for protecting private information and system authentication. Nevertheless, as the demands for ultra-low latency and high reliability have increased, the traditional security management models have been unable to meet these requirements without incurring significant operating costs and overhead [16]. Deployment of 5G and 6G in V2X communications will provide novel security capabilities that will compensate for the drawbacks of 802.11p, DSRC, and LTE-V2X. The 5G and 6G security models have facilitated the improvement of security in aspects of network programmability and flexibility while meeting the specific security requirements of every network user and providing stable Quality of Experience (QoE). Network Function Virtualization (NFV), Software-Defined Networking (SDN), and network slicing encompass the majority of technologies that enable highly innovative security mechanisms for 5G and 6G-based V2X communications. Moreover, in 5G, security services are provided in two stages, first via the architecture and second via supporting technologies like NFV and SDN [115], [163], [164]. 5G provides availability, data integrity, authentication, and confidentiality within its main architecture. User equipment and 5G network units, such as mobility management units and other service providers, natively support authentication services. This is the primary

difference between classical (3G and 4G) cellular networks and 5G. However, the ultra-fast and efficient authentication strategies in 5G should be considered in future related research. In addition to the benefits mentioned above, 5G improves data security and confidentiality by concentrating on the lower communication layers, which are susceptible to a set of known attacks and must be safeguarded against them. More precisely, 5G inherits data integrity services from the upper layers, even if the 5G core architecture is able to protect data integrity information at the lower layers. Through the use of Direct-Sequence Spread Spectrum (DSSS) and Frequency-Hopping Spread Spectrum (FHSS), 5G also alleviates the effects of attacks like jamming and DoS at the lower layers.

In [153], the authors study the possibilities of leveraging the 5G technology's ability to support higher data rates with a more significant number of connected objects in order to prevail over public-key cryptography challenges. The authors suggest a new approach for securing V2X communications in VANETs, which utilizes a registration list and elliptic-curve public-key cryptography. To protect against attacks such as DoS, message modification, and eavesdropping attacks, this method only needs two lightweight hash operational processes. The experimental findings show that the approach provides relatively low authentication latency even when the vehicle density is very high. Nevertheless, it depends on a non-realistic premise of reliable access points and wireless networks. According to work in [154], proactive security countermeasures such as cryptography techniques are susceptible to internal attacks (for example, position falsification and false warning generation attacks) performed by authenticated vehicles. To mitigate these attacks, the authors suggest a decentralized approach for detecting malicious behavior in 5G-based V2X communication and vehicular networks. The research work in [155] provides a comprehensive overview of recently enabled 5G-V2X slicing use cases and related security challenges focusing on cross-border slicing attacks. Moreover, this study introduces the open security issues of 5G-V2X slicing.

The authors of [156] introduce batch authentication and key exchange schemes to provide a high level of security by evading communication with malicious vehicle users. In this scenario, RSUs can simultaneously authenticate a group of vehicle users in a coverage region to relieve the computational load. Moreover, it is considered that 6G can give excellent transformations in VANETs that will improve it in comparison with the previous generations in terms of network capacity and high data rate. In [157], a model is proposed to maintain privacy and security in cloud data centers and data transmission during resource management and offloading. The proposed technique provides a system that is able to train a model and help update the edge devices' data. The whole edge cloud system will provide updated information for edge devices and will solve the problems of getting some important information necessary for model-related optimization. Moreover, it will improve the

efficiency of the frameworks of the 6G-V2X communication network. The authors of [158] explore the ways that 6G and the Internet of Autonomous Vehicles (IoAV) can cooperate to satisfy the applications of the next generation of connected and autonomous vehicles. Then, they propose 6G cellular communication to support the quality of the next IoAV applications and try to investigate the main security challenges in the 6G-IoAV.

V. OPEN ISSUES AND FUTURE RESEARCH DIRECTIONS

As a result of communication advancements, the upcoming vehicular networks will include all roadside entities, not only RSUs and vehicles. Therefore, V2X communication networks will include new security issues that traditional security strategies cannot appropriately face. In other words, various security concerns require additional consideration to develop more secure V2X communications. This section highlights some of these concerns.

A. OPEN ISSUES

Some of the most important open issues, mainly related to security challenges in vehicular networks and V2X communications, are listed as follows:

- The effectiveness and performance of security reporting response systems rely on the existing V2X data, which can be different in various network components and layers. However, such data can be inadequate to detect misbehaviors that are carried out in a distributed way. Under such circumstances, there is a propensity to gather distributed information into a single central unit and execute data-driven detection utilizing historical information. Typically, characteristics like velocity, geographic position, and density are used to cluster vehicles geographically. However, related to network availability, the servers of mobile edge computing and roadside units are sparsely deployed because of their high installation and operating expenses. Road traffic patterns and network coverage are also regarded as necessary prerequisites that should be taken into account for the installation of stationary infrastructure units. From an effective implementation standpoint, these constraints could affect the effectiveness of security strategies. Distributed data transmission to a central unit can, additionally, result in high consumption of resources and delay, which, consequently, can violate the performance requirements of V2X. Thus, the emerging challenges related to inadequate V2X data existence, security issues associated with V2X data transmission between vehicular network entities, and the problems related to the accessibility and reliability of the deployment of RSUs need to be investigated as potential open issues.
- RSUs are employed as completely trusted units in updating security information in almost all centralized security mechanisms. Therefore, vehicles must communicate with the RSUs to obtain updated information. However, since RSUs are sometimes not accessible

along the roadsides, updating information is always challenging in vehicular networks. Moreover, RSUs, same as other road entities, are susceptible to a variety of attacks. Therefore, challenging issues related to the reliability, accessibility, and security of RSUs are still some open issues that should be studied.

- A large percentage of security methods are verified via theoretical system evaluation and system-level simulations. Nevertheless, current simulators are almost weak and restricted in accurately representing the intricacy and dynamics of actual V2X communication networks. Additionally, the considerable majority of the investigations (provided so far) concentrate on particular attack instances and make specific presumptions in their theoretical systems that may not accurately represent the characteristics of V2X in real scenarios. On the other hand, most researchers do not make the used datasets in their evaluations publicly available, which reduces the likelihood of comparative feasibility investigations and repeatability, creating previously unheard-of difficulties in evaluating security solutions in real-world V2X networks. Therefore, performing more accurate simulations with real-world approximation, investigating security attacks on V2X communications where the investigation focuses on constraints that represent the characteristics of V2X in real scenarios, and constructing new and considerable datasets for V2X communication to perform novel learning-based research and finding effective V2X security solutions are important issues which should be considered.
- Information in V2X platforms can be distributed via multi-hop V2I and V2V communications, where DSRC units are responsible for rebroadcasting received data after receiving. Earlier studies in this field have demonstrated that a single RSU is capable of efficiently spreading a Certificate Revocation List (CRL) in an urban area. However, several security problems, such as the reliability of these systems and their capability to overcome network congestion, must be addressed.
- Wireless communication is a developing technology that has the potential to improve communication within public transportation systems like buses, trains, and metros. It is, however, more vulnerable to a variety of cyberattacks than wired communication. Therefore, in the context of V2X communications technology, securing wireless communication, as one of the main components of V2X communication structure, against security threats, especially cyber attacks such as Eavesdropping, Man-in-the-middle, and Evil twin attacks, is always a potential open challenge in the field of vehicular networks.
- It is self-evident that the cost of incorporating advanced V2X technologies into an automobile raises the vehicle's cost. Furthermore, the expense of upgrading cities and educating first-time purchasers adds to the total cost of implementing V2X technologies and attracting customers. However, security concerns should still be considered when implementing efficient and economical V2X communications technology. Consequently, finding novel techniques to implement secure, efficient, and economical V2X technology so that it is accessible to more customers is an important point of view.
- One of the most effective ways to reduce learning time and increase resource efficiency in large-scale V2X communication networks is utilizing edge components through distributed learning models and decentralized learning strategies. Through the dissemination of distributed intelligence within a V2X network, vehicles become intelligent, cooperative entities capable of carrying out cooperative and supportive learning tasks free of centralized management. Cooperative learning supports local decisions and enables vehicles to share locally learned and trained model settings and parameters. While FL-based strategies are making improvements toward privacy-preserving V2X systems, they continue to depend on a server-client structure, which leaves them susceptible to various malicious activities. Furthermore, adversarial strategies and data poisoning may affect locally trained ML-based models. Therefore, when novel strategies based on federated learning techniques are proposed, considering their robustness against malicious and adversarial activities is a critical issue still open to research.
- Vehicular communication networks are often multi-tenant, with various participants including network operators, road authorities, manufacturers of original equipment, municipalities, and service providers. Unlike former generations of mobile networks, the structure of 5G and 6G networks makes use of cloud-native technologies to enable a segmentation of network operations. Therefore, several logical networks may be hosted concurrently by 5G and 6G-enabled vehicle infrastructure, enabling multi-tenancy and multi-service implementation and deployments. Moreover, V2X communication services can be constructed in a customizable, flexible, and dynamic manner using software-defined networking, network functions virtualization, and cloud computing in accordance with the network-as-a-service paradigm. There are new end-to-end security and trust issues that arise in this particular scenario. Therefore, in addition to traditional perimeter protection strategies, the dynamic coordination of multi-tenant and multi-domain security rules is crucial. With these circumstances, distributed ledger technologies like blockchain may be used to provide reliable interoperability between the multiple participants engaged in the V2X communication service chain. However, this may introduce additional time complexity. Therefore, practical technological strategies focused on the speedup of encryption and decryption operations and, in general, on improving the scalability of distributed ledger solutions may alleviate the problem and is a potential open area to work.

B. FUTURE RESEARCH DIRECTIONS

This subsection suggests some possible future research directions based on the above-mentioned open issues related to V2X communications technology and its security challenges and countermeasures.

- The emerging challenges related to inadequate V2X data existence, security issues associated with V2X data transmission between vehicular network entities, and the problems related to the accessibility and reliability of the deployment of RSUs need to be investigated as potential open issues. Therefore, a relatively undiscovered research avenue for overcoming V2X implementation and deployment constraints is to use the physics of traffic through consistency checks versus basic physical rules provided by traffic flow theory.
- Since challenging issues related to the reliability, accessibility, and security of RSUs are still some open issues that should be studied, future research should solve these challenges or propose proactive solutions for avoiding dependencies on RSU reachability in security information updates.
- As mentioned, performing more accurate simulations with real-world approximation, investigating security attacks on V2X communications where the investigation focuses on constraints that represent the characteristics of V2X in real scenarios, and constructing new and considerable datasets for V2X communication to perform novel learning-based research and finding effective V2X security solutions are important issues which should be considered. Consequently, more research activities must be done to construct suitable and open-source datasets and make them accessible to the research and scientific communities.
- In some multi-hop V2X communications, DSRC units are responsible for rebroadcasting received data after receiving it. In these situations, security problems, such as the reliability of RSU systems and their capability to overcome network congestion, must be addressed. Therefore, working on these security issues in the given circumstances is a potential direction for future studies.
- Given the fact that in the context of V2X communications technology, securing wireless communication, as one of the main components of V2X communication structure, against security threats is always a potential open challenge, for future research, it is suggested that novel and robust security mechanisms for wireless communications be introduced, particularly in the case of vehicular communication networks.
- As mentioned before, it is essential to develop efficient techniques to implement secure, efficient, and economical V2X technology so that it is accessible to more customers. Accordingly, appropriate research efforts must be adopted to reduce the total cost of procurement and make future deployments of V2X technologies more efficient. This can be a possible future research direction.

- When novel strategies based on federated learning techniques are proposed, their robustness against malicious and adversarial activities must be considered. Thus, designing robust and privacy-preserving FL-based models that can mitigate these kinds of vulnerabilities is a potential research area.
- The traditional trust, privacy, and security strategies may be inadequate to meet the upcoming challenges associated with the integration of 5G and 6G technologies in V2X communications and vehicular networks. It seems necessary to do more studies focusing on the current conditions of 5G- and 6G-based V2X to develop and optimize adaptive security mechanisms. It will be a potential research direction for future studies.

Other directions for future work can include finding novel and effective solutions for protecting V2X communication systems against new message spoofing, man-in-the-middle, replay, DoS, and DDoS attacks, as well as emerging menaces that can affect these infrastructures. Moreover, threats to remote communication technologies (as mentioned before) and databases should be addressed in future research.

VI. CONCLUSION

In this survey, we have comprehensively reviewed existing representative research work on the architecture, main characteristics, and communication models of the V2X communication network. Based on the critical analysis of the existing research trends, we have mentioned some of the most significant applications of these vehicular communication networks and also we analyzed the potential threats and vulnerabilities for V2X advanced technologies and presented the existing security countermeasures against these threats. Moreover, we categorized the current solutions based on the security strategy used to implement them. Eventually, we mentioned some of the open issues and future research directions in this field. This article has examined the studies of the last ten years and compiled them. We hope that this survey will help make it easier for researchers studying V2X communication subjects and related fields to keep track of the academic frontier, by offering a landscape and a roadmap.

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